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Industrial distribution and its effect on land surface temperature (LST) in Sukoharjo Regency in 2001, 2009 and 2021

Fitria Dewi Kartika*, Rika Kurnia Dewi and Pipit Wijayanti

Magister Geography Education, Universitas Sebelas Maret, Indonesia

*Email : fitriadewi27@student.uns.ac.id

Abstract. The expansion of human activities, driven by population growth and various societal needs, has led to significant land-use changes, resulting in microclimate modifications. These changes have a crucial impact on land surface temperature (LST), particularly in urban areas. Urbanization, transportation, industrialization, and urban settlements are the main factors contributing to LST variations. This study focuses on the distribution of industries and their influence on land surface temperature in Sukoharjo Regency, Central Java, during the years 2001, 2009, and 2021. The research utilizes Landsat satellite imagery to analyze the industrial distribution and LST in the study area. The images were acquired on August 27, 2001, June 14, 2009, and July 25, 2021. The methodology includes land-use classification, LST extraction, and regression analysis to assess the relationship between industrial land area and land surface temperature. The results show a significant increase in the number of industries in Sukoharjo Regency over the studied period, with a growth rate of over 100% from 2001 to 2010 and further expansion until 2021. The industrial areas exhibited a clear spatial distribution in all years, indicating the impact of industrialization on land use. The regression analysis revealed a relationship between the industrial land area and land surface temperature, suggesting that expanding industries contributes to higher land surface temperatures. These findings provide valuable insights for urban planning and environmental management in Sukoharjo Regency to mitigate the impacts of climate change and promote sustainable development.

1. Introduction

Human activities are many and varied; considering the rise in global population, there is a need for more settlements, roads, an increase in agricultural production, an increase in demand for food and shelter, a change in lifestyle, etc., which invariably leads to increased encroachment on natural vegetation [1]. This change in land use causes changes in microclimate. Changes in microclimate are very important because humans live in a microclimate [2]. Thus, built-up areas tend to have a higher land surface temperature (LST) compared to the surrounding suburban environment [3,4]. Urbanization, transportation, industry, and housing are the main activities potentially related to LST [5,6]. In other words, population growth in urban areas is increasing exponentially, both naturally and as a result of urbanization, giving rise to natural problems, namely Land Use or Land Cover which is replaced, giving rise to other problems such as changes in surface temperature. . This is because changes in land use affect changes in the wavelength of solar radiation that reaches the earth's surface and is reflected back into space.

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As is the case in Sukoharjo Regency, many land changes have been replaced by industrial development. The number of industries in the Sukoharjo Regency area continues to increaseearly. Medium and large industries in Sukoharjo Regency from 2002-2006, respectively, grew at ,1.93%, ,6.57%, 20.77%, and 5.99%. Moreover, there are regulations regarding industrial development, Sukoharjo Regency Regional Regulation (Regional Regulation) Number 35 of 2002 concerning the Regional Development Program (Propeda) of Sukoharjo Regency for 2001-2006. As a result of regulations regarding industry, many industries have developed in the Sukoharjo Regency area. Based on sectoral contributions to Sukoharjo Regency's GRDP imost significant the largest contribution was from the industrial sector. Based on Sukoharjo Regency Regional Regulation Number 2 of 2018 concerning industrial development plans in Sukoharjo Regency for 2018-2038, the industrial allotment area in Sukoharjo Regency itself is 2,352 ha where the industrial allotment area is in the sub-districts: Kartasura, Grogol, Sukoharjo, Gatak, Ngute, Bendosari, Tawangsari, Polokarto and Mojolaban, while industrial zones will be directed to industrial designated areas, namely in Nguter and Bendosari sub-districts.

Urban land surface temperature can be detected through remote sensing. the advent of remote sensing, using various satellite and airborne sensors that register thermal infrared information it became possible to achieve calculations of land surface temperature (LST) [7]. Satellite imagery, especially Landsat imagery, has the ability to detect vegetation density and surface temperature [8]. Landsat ETM+ with 60 m, spatial resolution of the thermal infrared band allows experts to determine more detailed surface temperatures [9]. Changes in surface temperature will be visible if identified temporally. Thus, more than one Landsat image must be taken with different observation times. This research took three years, namely 2001 (before regulations), 2009 (after regulations), and 2021 (last year). Based on the background that has been stated, it is necessary to conduct research on the distribution of industry and its impact on land surface temperatures in Sukoharjo Regency in 2001, 2009 and 2021.

2. Methods

The research location is in Sukoharjo Regency, Central Java Province. The data used in this research are Landsat 7 ETM+ Imagery data with an acquisition date of 27 August 2001, Landsat 7 ETM+ Imagery with an acquisition date of 14 June 2009 and Landsat 8 OLI/TIRS Imagery with an acquisition date of 25 July 2021. Regarding the acquisition date, the image scene was chosen for held in the same season and on dates that are not too far apart, to eliminate differences in solar illumination and differences in vegetation and soil conditions [10]. Data collection was carried out by field observation to test image accuracy and carry out secondary data analysis. Meanwhile, the analysis technique used is as follows:

2.1. Analysis of industrial distribution in Sukoharjo Regency

Industrial distribution was obtained through land use maps from Landsat Image analysis in 2001, 2009 and 2021. Industrial distribution in Sukoharjo Regency was collected in the form of industrial coordinates in Sukoharjo Regency using the Global Positioning System (GPS). Then, the plotting results are processed to obtain a distribution map in Sukoharjo Regency. The image that has been processed is then tested for image accuracy. Categorized pixels from the image are compared to the same site in the field. Accuracy assessment results typically provide the user with overall map accuracy and accuracy for each class on the map [11].

2.2. Analysis of land surface temperature in Sukoharjo Regency

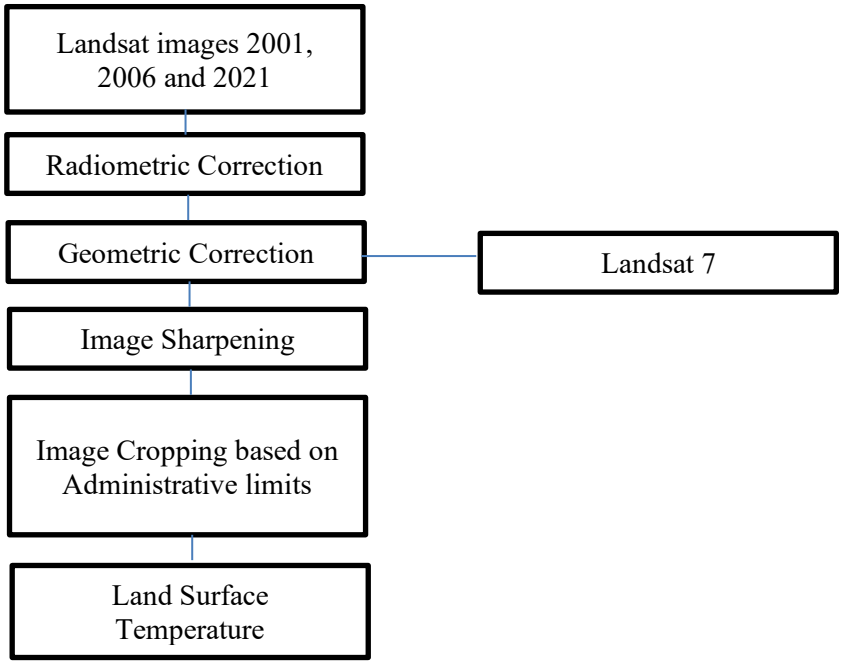


Figure 1. Image Processing Flowchart for Land Surface Temperature (LST) Analysis

To find out LST data extraction from Landsat thermal bands must go through six different steps using ENVI 5.3 and ArcGIS 10.8 software. Following are the steps [12,13] :

2.2.1. TOA beam conversion. Every object in the world reflects some thermal electromagnetic energy because its temperature is more than 0 (zero) K, which is known as absolute 0 (zero). Using this principle, the thermal sensor (ETM+) receives a signal that can be converted into an atsensor emission. Calculation of spectral radiance (Lλ) has been carried out using the following equation:

$$L\lambda = ML * QCAL + AL \quad (1)$$

2.2.2. Convert to spectral radiance to Satellite Brightness Temperature

$$T = \frac{K2}{In\left(\frac{K1}{L\lambda}+1\right)} \quad (2)$$

Information
T = At-Satellite Brightness Temperature
Lλ = TOA Spectral Radiance
K_1 = Constant Band
K_2 = Constant Band

2.2.3. Vegetation Proportion (Pv) can be calculated

$$Pv = ((NDVI - NDVI_{min})/(NDVI_{max} - NDVI_{min}))^2 \quad (3)$$

2.2.4. The ground surface emissivity (e) can be obtained

$$E = 0{,}004 * Pv + 0{,}968 \quad (4)$$

2.2.5. Land surface temperature (LST)

$$BT/1 + W * (BT/P) * Ln(e) \quad (5)$$

Information
BT = At-satellite Brightness Temperatures
W = Wavelegth of emitted radiance

2.2.6. Convert LST data to Celsius scale. Convert this Kelvin (K) data to Celsius (C) for simplified and widely understandable temperature data.

2.3. Analysis of the influence of industry area on surface temperature
The data analysis technique used to determine the influence is Regression Analysis, by finding whether there is an influence between variables. To find out the significance of the regression equation, you need to compare the empirical f values with the theoretical f values, which are found in the F value table. Whether there is a significant influence of variable X on variable Y, then the calculation results are compared with a significance level of 5%. If F count > F table, then there is an influence of variable X on variable Y.

3. Results and Discussion

The images used in the Industrial Distribution and LST analysis are Landsat 7 and Landsat 8 images downloaded at https://glovis.usgs.gov/. The selected image has a recording time that is not too far apart. In 2001, the data used was Landsat 7 imagery with a recording date of 27 August 2001; in 2009, using Landsat 7, the recording date was 14 June 2009; and in 2021, using Landsat 8 imagery with a recording date of 25 July 2021. However, the 2009 Landsat 7 image was damaged. So there is a gap. The three images selected have the lowest cloud cover, namely less than 5%, so distortion due to clouds is minimal. The image is then processed in the Envi application to detect industrial land use and LST. The Landsat image processing results produce the industry and land surface temperature distribution.

3.1. Industrial Distribution
Landsat images are processed to obtain industrial and non-industrial land cover to determine industry distribution. Industry can be identified by identifying groups of pixels with similar elements such as hue, colour, texture and other elements. Industrial has a white/silver colour, bright hue, smooth texture, regular pattern, and square shape and is associated with the road. Once the calcification results are known, an image accuracy test is carried out. Image accuracy test results show an overall accuracy value of 88.8% in 2001, 94.1% in 2009 and 93.01% in 2021.

In 2001, it was recorded that there were 19 large industries in Sukoharjo Regency [14], in 2010 there were 58 industries [15] and as many as 204 industries in 2021. This shows that Sukoharjo Regency experienced an increase in numbers of more than 100% in the period 2001 to 2010 and until 2021.

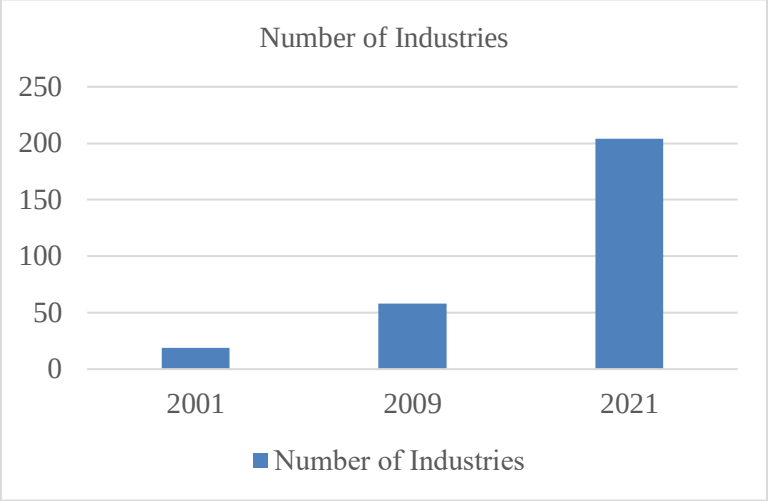


Figure 2. Graph of Number of Industries in 2001, 2009 and 2021

The results of the interpretation of Landsat 7 images in 2001 and 2009 and Landsat 8 in 2021 show that the number of industries is not visible, but their distribution is apparent. From the results of the interpretation, it can also be found that the area of industrial objects from 2001, 2009 to 2010 has increased rapidly. This is supported by the Sukoharjo Regency Regional Regulation (Regional Regulation) Number 35 of 2002 concerning the Regional Development Program of Sukoharjo Regency for 2001-2006, which makes it easier for industry to develop in Sukoharjo Regency. The industrial land area in 2001 was 161.05 Ha, then increased by 50.4% in 2009 to 242 Ha. From 2009 to 2021, it rapidly increased, reaching 218% from 242 Ha to 772 Ha.

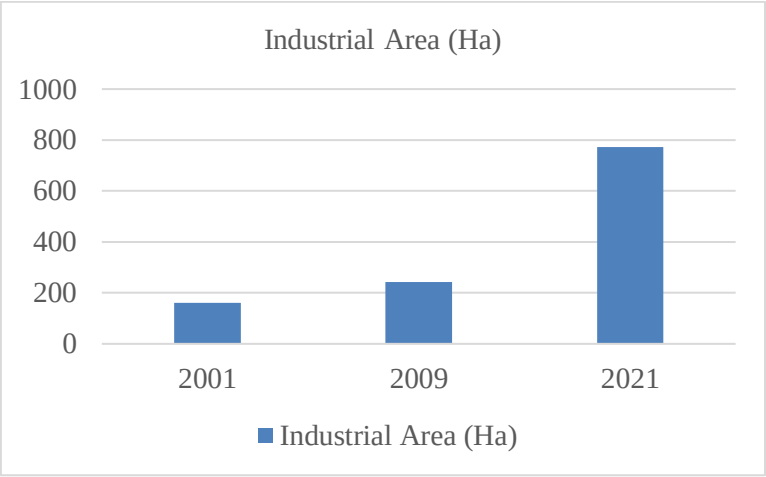


Figure 3. Industry Area Graph for 2001, 2009 and 2021

In 2001, industrial land was spread across Sukoharjo District, Grogol District and Kartasura District. In 2009, industrial land began to expand in other sub-districts, namely Gatak and Nguter. In line with government policy to support industrial development and investment, in 2021, all industries will be located in all sub-districts in Sukoharjo Regency. However, the industry will remain focused in Grogol Sub-district, with the most significant industrial area compared to other sub-districts. The industry distribution in Sukoharjo Regency is always associated with main roads because of its proximity to industrial accessibility. The pattern of industrial distribution is concentrated in areas directly bordering Surakarta City, namely Grogol District and Kartasura District, so it can be seen on

the industrial distribution map that the industrial land use detected is very dense in these two sub-districts.

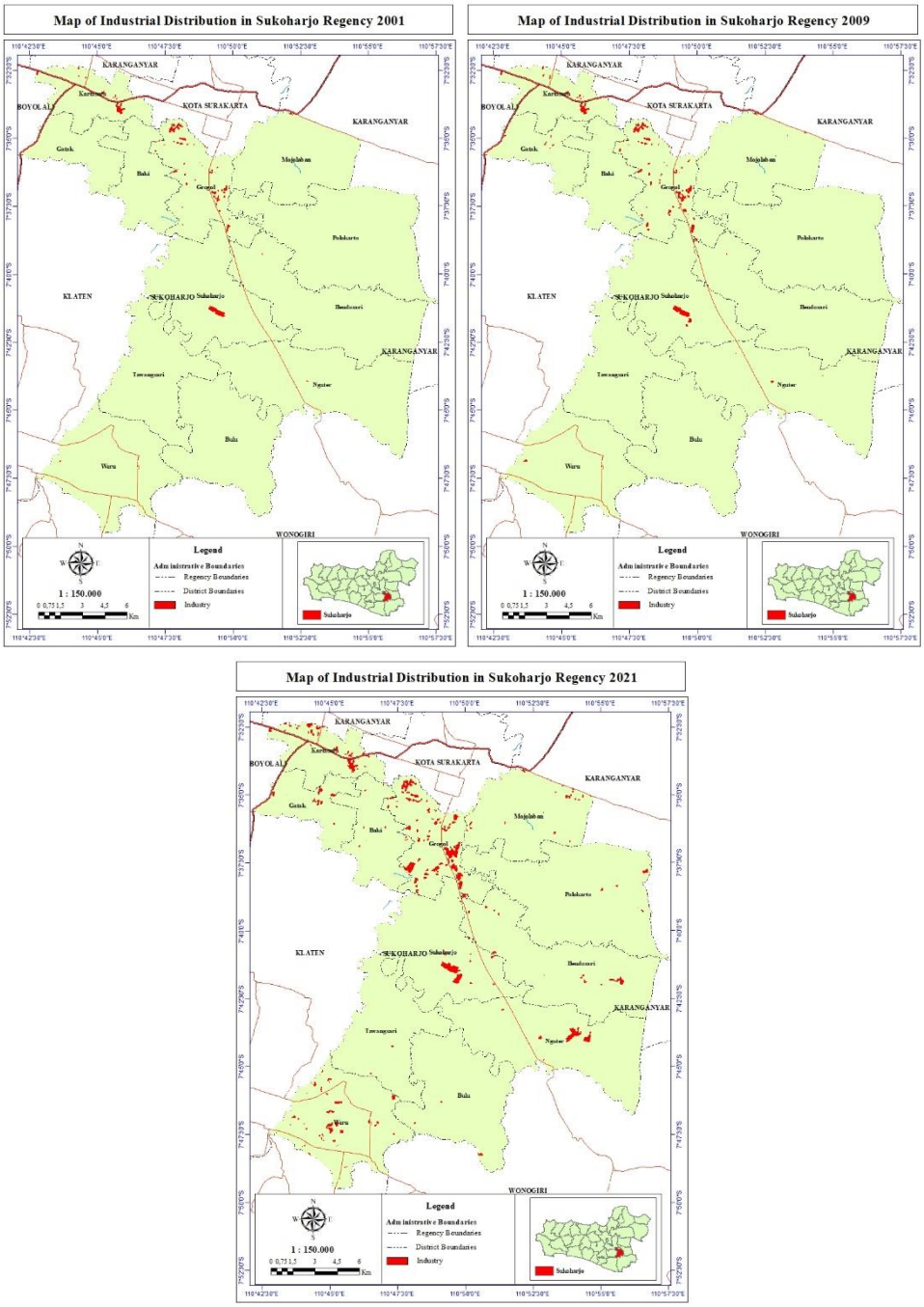


Figure 4. Industry Distribution Map in 2001, 2009 and 2021

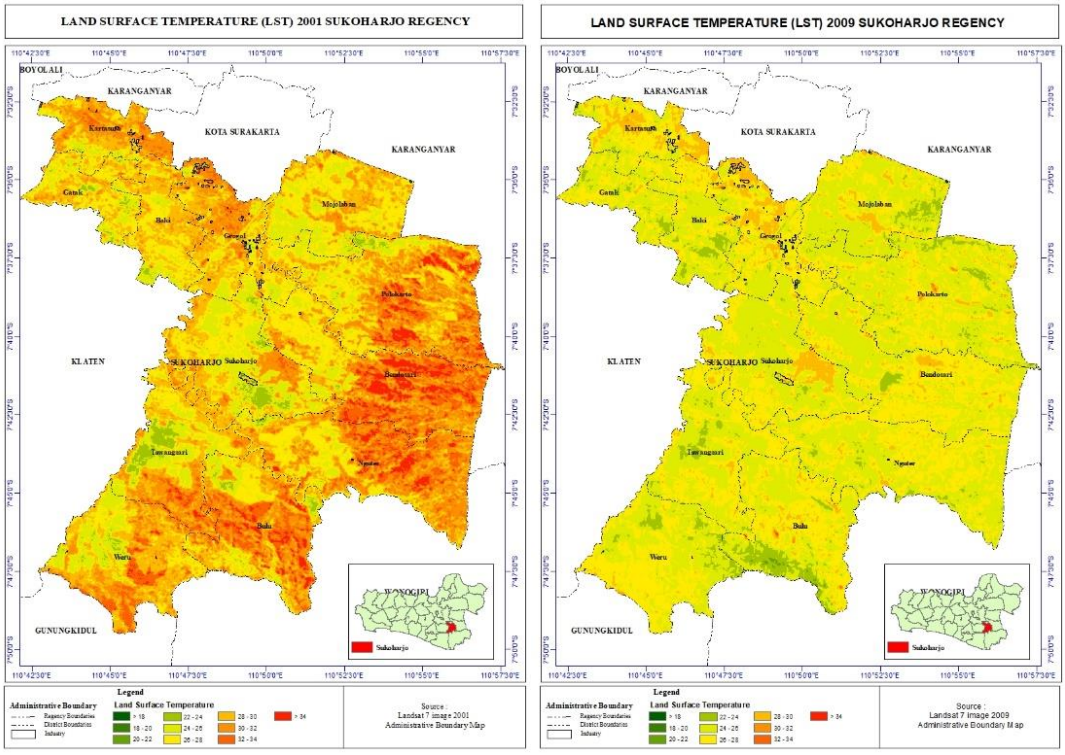
3.2. Land Surface Temperature

The results of Landsat image processing show that land surface temperatures decreased from 2001 to 2009 and increased in 2021. In 2001, the dominant surface temperature was 28 ˚C – 30 ˚C; in 2009, the dominant surface temperature was 24 ˚C - 26 ˚C, while in 2021, the dominant surface temperature will be a temperature in the range of 28 ˚C – 30 ˚C. Land surface temperature in 2001 had the highest temperature distribution in the hilly area bordering Wonogiri Regency because, at the time the

recording was carried out during the dry season, the dominant vegetation was teak trees, so it seemed as if the area had no vegetation and was very dry. The ground surface temperature has increased on solid surface objects in urban areas. The phenomenon of decreasing temperature occurs in some industrial land uses. Large industrial areas allow for variations in surface temperature in an industry.

Table 1. Land Surface Temperature Area in the industrial area of Sukoharjo Regency

Temperature (˚C)	Area (Ha)		
	2001	2009	2021
>18	0,72	0,63	0,63
18 - 20	1,80	1,44	1,71
20 - 22	6,03	6,93	8,55
22 - 24	13,41	29,13	49,02
24 - 26	26,37	71,20	260,22
26 - 28	37,44	91,69	369,85
28 - 30	36,68	36,09	74,46
30 - 32	26,12	3,60	5,49
32 - 34	9,24	1,26	1,53
> 34	3,24	0,27	0,27
Total Industrial Area	**161,05**	**242,24**	**771,73**



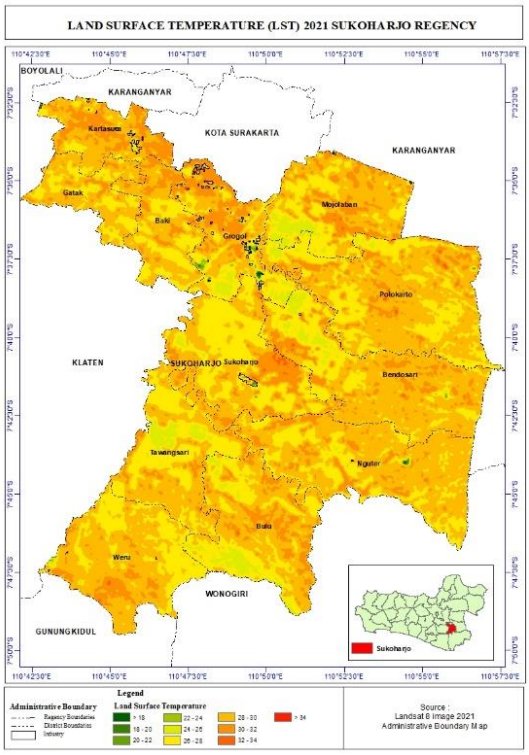


Figure 5. Land Surface Temperature Map of Sukoharjo Regency

The results of Landsat Image Processing for 2001, 2009 and 2021 show that several industrial points have decreased temperature levels due to land conversion, for example, in the industry (point In 2001, the region had a temperature in the range of 30˚ C- 32˚C, but this decreased to a temperature in the range of 26 ˚C-28 ˚C in 2009 and decreased again in 2021 with a temperature range of 24 ˚C-26 ˚C. The following is a sample graph of industries that experienced increases and decreases in temperature:

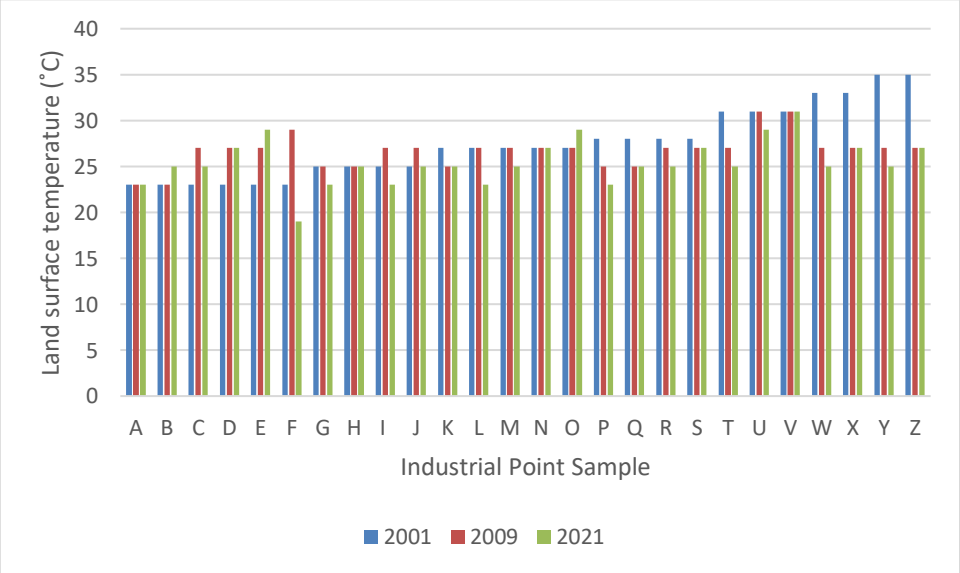


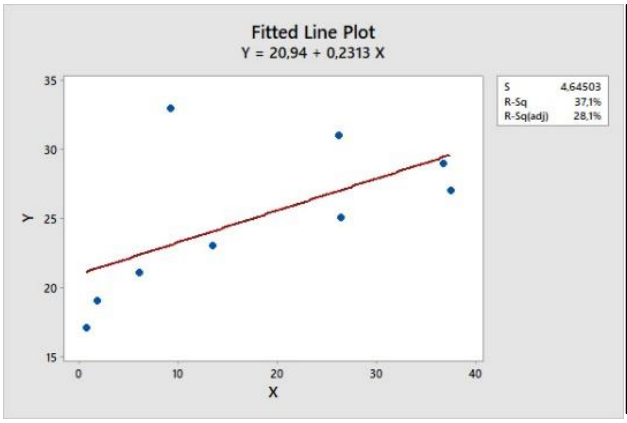
Figure 6. Industry Sample Graph

Of the 26 industrial sample points observed, 4 sample points showed a constant temperature, 4 sample points showed an increase in temperature, and 18 sample points experienced a decrease in temperature. This shows that land surface temperature does not necessarily show a positive relationship; surface temperature will increase if industry increases, but land surface temperature can also decrease. This decrease in 2009 was possible because the land surface temperature analysis results were less than optimal. After all, the image had an error, so the recording of areas experiencing gaps would be generalized from neighbouring areas. The decrease in temperature in 2021 occurred due to production restrictions during the pandemic, so land surface temperatures decreased. Time series changes in LST can reflect the impact of the COVID-19 pandemic on industrial production and indicate pollution in the surrounding region [16,17].

3.3. Industry-wide influence on surface temperature

Land changes in the Sukoharjo Regency area continue to increase, mainly in industrial areas. Land changes continue to increase because they are in the development stage, so land changes increase. Increasing changes to industrial areas influence surface temperatures; this is shown by the results that surface temperatures in 2001, 2009 and 2021 continue to increase. Linear regression was carried out between the industrial area and surface temperature to determine the relationship, where x is the industrial area, and y is the land surface temperature. The regression results can be seen as follows:

3.3.1. The relationship between industrial area and soil temperature in 2001.



Regression Analysis: Y versus X
The regression equation is
Y = 20,94 + 0,2313 X
Model Summary

S	R-sq	R-sq(adj)
4,64503	37,07%	28,08%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	88,966	88,9657	4,12	0,082
Error	7	151,034	21,5763		
Total	8	240,000			

Figure 7. Regression results of industrial area (X) and land surface temperature (Y) in 2001

Based on the regression results, it can be seen that the p-value is 0.082, which means more than 0.05, so the relationship between the two has an influence but is very small. The variation or result of surface area in 2001 is explained by the industrial area of 37.7%, which means that the industrial area in 2001 had an influence of 37% on the land surface area in 2001; this shows that industrial area is not the only factors that explain the surface area of the land. The coefficient of the industrial area variable is positive, which means that every increase in the value of the industrial area variable (X) by 1 unit will increase the temperature (Y) in Sukoharjo Regency by 0.2313, assuming other variables are constant.

3.3.2. Relationship between industrial area and land surface temperature in 2009

Regression Analysis: Y versus X

Regression Equation

Y = 19,89 + 0,0918 X

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	1	64,49	64,493	6,79	0,048
X	1	64,49	64,493	6,79	0,048
Error	5	47,51	9,501		
Total	6	112,00			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
3,08243	57,58%	49,10%	28,76%

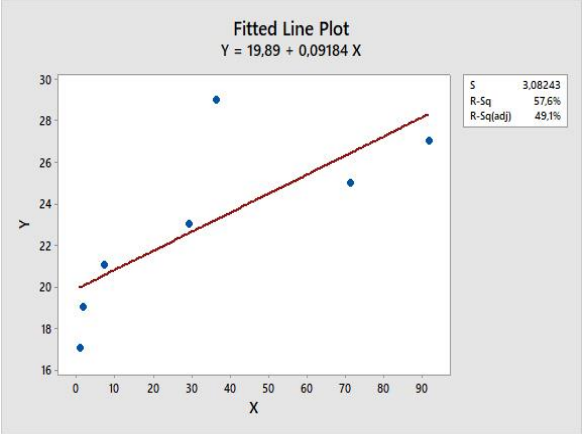
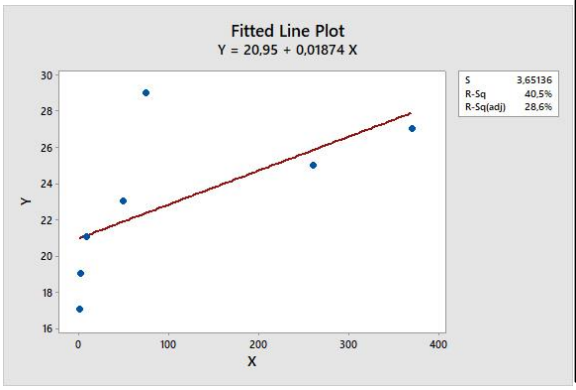


Figure 8. Regression results of industrial area (X) and land surface temperature (Y) in 2009

Based on the regression results, it can be seen that the p-value is 0.048, which means less than 0.05, so the relationship between the two has a significant influence because the p-value is <0.05. The variation or result of surface area in 2009 is explained by the industrial area of 57.58%, which means that the industrial area in 2009 had an influence of 57% on the land surface area in 2009; this increased from 2001, which was only 37% so that shows that industrial area is an essential factor influencing land surface temperature. However, industrial area is not the only factor that explains land surface area; other factors influence it by 43%. The coefficient of the industrial area variable is positive, which means that every increase in the value of the industrial area variable (X) by 1 unit will increase the temperature (Y) in Sukoharjo Regency by 0.09, assuming other variables are constant.

3.3.3. The relationship between industrial area and land surface temperature in 2021



Regression Analysis: Y versus X

The regression equation is

Y = 20,95 + 0,01874 X

Model Summary

S	R-sq	R-sq(adj)
3,65136	40,48%	28,58%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	45,338	45,3379	3,40	0,124
Error	5	66,662	13,3324		
Total	6	112,000			

Figure 9. Regression results of industrial area (X) and land surface temperature (Y) in 2021

Based on the regression results, it can be seen that the p-value is 0.124, which means more than 0.05, so the relationship between the two has an influence but is very small. The variation or result of surface area in 2021 is explained by the industrial area of 40.48%, which means that the industrial area in 2021 has an influence of 40% on the land surface area in 2009; this shows that the industrial area is not the only factors that explain the surface area of the land. The coefficient of the industrial area variable is positive, which means that every increase in the value of the industrial area variable (X) by

1 unit will increase the temperature (Y) in Sukoharjo Regency by 0.018, assuming other variables are constant.

The regression calculations for 2001, 2009 and 2021 show that industrial area can influence land surface area, but this is only one factor that can influence it because other land covers can influence the increase in land surface temperature. The coefficient for the industrial area variable in 2001, 2009 and 2021 has a positive coefficient, which means that for every increase in industrial area (X) in Sukoharjo Regency by one unit, the temperature (Y) in Sukoharjo Regency will increase with the assumption that other variables are constant.

4. Conclusions

The results of the research have conclusions that answer the problem formulation. The following are the conclusions in this research:

a. In 2001, it was recorded that there were 19 large industries in Sukoharjo Regency [14], in 2010 there were 58 industries [15] and as many as 204 industries in 2021. This shows that Sukoharjo Regency experienced an increase in numbers of more than 100% in the period 2001 to 2010 and until 2021. The distribution of industries in Sukoharjo Regency is always associated with main roads because this is related to proximity to industrial accessibility. The industrial distribution pattern is concentrated in areas directly bordering Surakarta City, namely Grogol District and Kartasura District.
b. The results of Landsat image processing show that land surface temperatures decreased from 2001 to 2009, and increased in 2021. In 2001 the dominant surface temperature was 28 ˚C – 30 ˚C, in 2009 the dominant surface temperature was 24 ˚C - 26 ˚C, while in 2021 the dominant surface temperature will be a temperature in the range of 28 ˚C – 30 ˚C. Of the 26 industrial sample points observed, 4 sample points showed a constant temperature, 4 sample points showed an increase in temperature and 18 sample points experienced a decrease in temperature. This shows that land surface temperature does not necessarily show a positive relationship, in the sense that surface temperature will increase if industry also increases, but land surface temperature can also decrease.
c. The results of Landsat image processing show that land surface temperatures decreased from 2001 to 2009, and increased in 2021. In 2001 the dominant surface temperature was 28 ˚C – 30 ˚C, in 2009 the dominant surface temperature was 24 ˚C - 26 ˚C, while in 2021 the dominant surface temperature will be a temperature in the range of 28 ˚C – 30 ˚C. Of the 26 industrial sample points observed, 4 sample points showed a constant temperature, 4 sample points showed an increase in temperature and 18 sample points experienced a decrease in temperature. This shows that land surface temperature does not necessarily show a positive relationship, in the sense that surface temperature will increase if industry also increases, but land surface temperature can also decrease.

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